IT4010 Research Project – 2025 July
Topic Assessment Form

Project ID : 25-26J-364

1. Topic (12 words max)

An AI-Powered Web Platform for Code Concept Extraction, Research Validation, Skill Assessment, and Interview Preparation for University Students

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

University students often face significant challenges in translating theoretical knowledge into practical industry-level skills. While academic programs typically emphasize programming fundamentals, data structures, object-oriented principles, and software engineering concepts, students are rarely equipped with tools to evaluate how well they understand and apply these concepts in real-world settings [1]. This disconnection between academic knowledge and industry requirements creates hurdles when students face technical interviews, develop software, or conduct original research.

One of the core issues is the absence of structured feedback mechanisms that guide students on their conceptual weaknesses and strengths. Popular platforms like LeetCode and HackerRank allow coding practice but do not provide insight into the underlying theoretical concepts being used, nor do they offer concept-level feedback or structured knowledge extraction from code [2].

Another significant problem is the lack of early-stage research validation tools. Many students, especially undergraduates, unknowingly pursue research topics that are not novel. Without access to systems that cross-reference existing scholarly work such as IEEE Xplore, students may end up replicating existing studies or submitting ideas that lack innovation [3]. Current plagiarism detection tools such as Turnitin or Grammarly primarily detect text-level duplication, not conceptual similarity or topic novelty. Hence, the need for a more intelligent semantic validation system is evident.

Furthermore, skill assessment platforms are often one-dimensional—offering either theoretical MCQs or practical coding but rarely combining both. They also lack adaptive features that recommend personalized learning paths or career options based on the student's performance and interests [4]. As a result, students struggle to map their current skills to suitable roles or upskilling paths.

Lastly, mock interview platforms are either too generic or expensive, offering static questions with no personalization or feedback. This leads to ineffective preparation for technical interviews, especially for students without access to mentors or career coaches.

Therefore, our research aims to develop an AI-powered educational web platform that unifies four essential components—code concept extraction, research validation, skill assessment with recommendations, and AI-powered interview simulation. The solution will provide automated, intelligent support throughout a student's academic-to-industry transition journey. It will help students understand the *why* behind the *what* they code, validate their research direction, prepare effectively for technical evaluations, and identify their best-fit career paths using artificial intelligence and semantic understanding.

References

[1] Robins, A., Rountree, J., & Rountree, N. (2003). *Learning and teaching programming: A review and discussion*. Computer Science Education, 13(2), 137–172.
https://doi.org/10.1076/csed.13.2.137.14200

[2] Ihantola, P., Ahadi, A., Karavirta, V., & Hellas, A. (2015). *Review of recent systems for automatic assessment of programming assignments*. ACM Transactions on Computing Education (TOCE), 15(2), 1–31.
https://doi.org/10.1145/2729380

[3] Hien, N. T., Dung, N. Q., & Ha, D. V. (2021). *A study of semantic similarity in scientific research topic recommendation using NLP techniques*. Journal of Intelligent & Fuzzy Systems, 40(2), 2711–2721.
https://doi.org/10.3233/JIFS-189310

[4] Ferguson, R. (2012). *The state of learning analytics in 2012: A review and future challenges*. Technical report KMI-12-01, Knowledge Media Institute, The Open University, UK.
https://kmi.open.ac.uk/publications/pdf/kmi-12-01.pdf

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

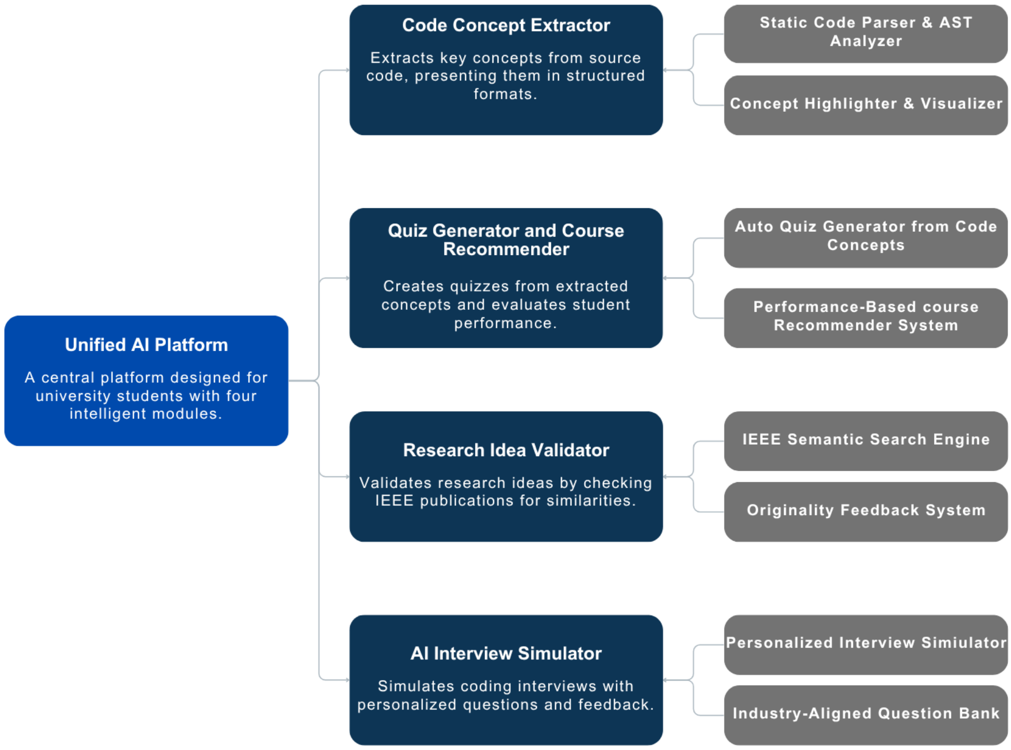

Figure I Conceptual Diagram

Our solution is a unified web-based AI-powered platform that integrates four intelligent modules, each targeting a specific pain point faced by university students:

1. **Code Concept Extractor** – Uses static analysis and NLP models to identify theoretical concepts from source code (like OOP, DSA, design patterns) and presents them in structured formats such as summaries, diagrams, or tables.
2. **Research Idea Validator** – Cross-references student-submitted research ideas with IEEE publications using NLP and semantic similarity checks. If similar topics are found, it suggests alterations to improve uniqueness and innovation.
3. **Quiz Generator and Course Recommender** – Creates quizzes from extracted concepts. Based on performance, it recommends personalized online courses from platforms like Coursera or Udemy.
4. **AI-Powered Interview Simulator** – Generates customized coding and theoretical interview questions based on user history and trending job requirements. After each session, detailed feedback and suggestions are provided.

All modules interact with a central user profile and learning database to deliver a personalized experience. The system is built with modern web technologies, integrated with AI/NLP libraries , and uses vector similarity search for semantic analysis.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Developing this platform requires expertise across several specialized domains within Information Technology:

1. **Natural Language Processing (NLP):** Essential for extracting programming concepts from code, evaluating research topic originality, and generating personalized quizzes and interview questions. Pretrained models such as BERT, GPT, or custom fine-tuned classifiers will be utilized for text understanding and generation.
2. **Static Code Analysis and Compiler Theory:** Required to analyze source code structure using abstract syntax trees (ASTs) and recognize patterns like object-oriented principles, data structures, and algorithms. This enables conceptual extraction from languages such as Java and Python.
3. **Information Retrieval and Semantic Search:** Crucial for the research validation module. Techniques such as vector embedding, cosine similarity, and semantic indexing will be used to compare student research ideas against IEEE publications, helping to determine novelty and relevance.
4. **Machine Learning for Adaptive Learning:** Powers the quiz and recommender modules. User performance data is used to personalize content, recommend suitable online courses, and suggest career paths aligned with individual progress and strengths.
5. **Web Development and UX Design:** Vital for delivering an intuitive and modular interface. Technologies such as React, Node.js, and relevant APIs will be used to create an interactive experience for quiz-taking, interview simulation, and result visualization.

Data Requirements:

- Student code samples for training/testing the Code Concept Extractor.
- Access to IEEE research abstracts for validating research ideas (via APIs or academic scraping).
- A curated question bank and job-role dataset for assessments and career recommendations.
- Real-world interview templates from open-source databases to generate realistic interview simulations.

This multidisciplinary integration ensures a robust, AI-driven platform that addresses key educational gaps for IT undergraduates.

8. Objectives and Novelty

Main Objective : To develop an AI-based integrated platform that supports code concept understanding, research idea validation, skill-based learning recommendations, and interview preparation for university students.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Himansha L.M.H IT22601360	Build the Code Concept Extractor to identify OOP, DSA, and design patterns from code.	• Parse user-submitted source code. • Use AST and NLP to extract and explain concepts. • Display concepts via diagrams, tables, or summaries.	Unlike standard code linters or syntax checkers, this tool focuses on theoretical understanding and visual presentation of concepts to enhance academic learning.
G.V.G.A.S.Hettiarachchi IT22604194	Implement the Quiz Generator and Course Recommender.	• Create quizzes from extracted concepts. • Track performance and match with	Dynamic and personalized quiz generation with career mapping based on coding theory not

		relevant learning resources.	commonly integrated in student tools.
Sandamal R.T IT22606860	Detect inefficient or error-prone code patterns suggest best-practice refactors provide explainable feedback with risk assessment and enable real-time interactive guidance.	• Parse code using AST • identify suboptimal patterns with AI/static analysis • generate refactor suggestions, assess risks, explain reasoning, and support user interaction for customized advice.	Combines explainable AI with risk-aware refactoring, offers an interactive chat-based refactor mentor, provides risk scoring for safer changes, and promotes developer learning alongside code improvement.
M.H.N.V.Dharmasiri IT22639226	Design the AI Interview Simulator.	• Generate questions based on user's previous activities. • Suggest Job recommendations for user's performance.	Personalized interviews based on concept history and performance more targeted than traditional mock interviews.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Himansha L.M.H IT22601360	The **Code Concept Extractor** component aligns strongly with the Information Technology specialization, particularly in areas such as **programming fundamentals, code analysis, data representation, and educational technology**. The Code Concept Extractor utilizes static analysis and natural language processing to parse and interpret user-submitted code. It identifies key academic concepts like object-oriented programming, data structures, and algorithm usage. These are presented visually through diagrams, tables, and summaries, enhancing

	understanding. This promotes learning effectiveness, one of the key goals in IT education. The system also supports IT-related outcomes like improving problem-solving ability, automating educational feedback, and building intelligent tutoring systems. It demonstrates practical application of programming, data modeling, and human-computer interaction—all core areas in IT.
G.V.G.A.S.Hettiarachchi IT22604194	The **Quiz Generator and Course Recommender** component is directly aligned with **IT applications in educational technology, adaptive systems, and career development tools**. The system uses the extracted programming concepts to generate customized quizzes and assess the user's understanding. Based on quiz performance and learning history, it recommends online learning paths and suitable job roles. This aligns with IT areas such as human-centered computing, intelligent systems, and digital education. The recommender system uses basic machine learning principles and user data profiling—core topics in IT—to personalize the experience and guide students in skill development and employability.
Sandamal R.T IT22606860	The **Code Refactor component** is closely aligned with core Information Technology principles, particularly in software engineering, programming languages, and intelligent software tools. Using static code analysis and AI-driven pattern recognition, this component not only detects and helps **fix common coding errors** but also **recommends best coding practices** to improve code quality. By suggesting cleaner, more efficient, and maintainable code structures, it enhances both **code efficiency** and **readability**. The system provides explainable feedback to guide developers in understanding the reasoning behind each refactor, fostering improved coding habits and adherence to industry standards. This component supports essential IT skills such as debugging, software optimization, and clean coding, making it valuable for both educational environments and professional software development.

M.H.N.V.Dharmasiri IT22639226	The **AI-Powered Interview Simulator** falls under the **AI, natural language processing, system integration, and professional readiness** domains of Information Technology. It mimics a real technical interview scenario, asks tailored coding and theory questions based on the user's progress, and gives personalized feedback. This module applies IT principles by creating an intelligent, user-adaptive interface. It uses NLP, dynamic content delivery, and feedback analytics to improve user performance. The component emphasizes soft skill development, career guidance, and system usability—relevant IT themes that prepare students for real-world applications in professional environments.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Sanjeevi	Chandrasiri	
Co-Supervisor	Ms.	Lokesha	Weerasinghe	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.